EXPERIMENTS

S.NO	NAME OF THE EXPERIMENTS
01	Determination of wavelength of given laser
02	Determination of acceptance angle and numerical aperture of an optical fibre
03	Determination of velocity of sound and compressibility of liquid using Ultrasonic interferometer
04	Determination of thermal conductivity of bad conductor using Lee’s disc method
05	Determination of viscosity of given liquid using Poiseuille’s flow method

SAMPLE VIVA QUESTIONS

1. What is an acronym for LASER?

   LASER stands for Light Amplification by Stimulated Emission of Radiation.

2. What is LASER?

   LASER is a device which produces a highly intense monochromatic beam of light which is coherent in nature.

3. What are the properties of LASER?

   (i) Highly intense,

   (ii) Coherent,

   (iii) Directional, Monochromatic, and

   (iv) Less divergent.

4. What is Stimulated emission?

   The process in which an atom an atom in the excited state returns to the ground state with an emission of emission of two photons by external stimulation.

5. What is the type of laser beam used in the experiment?

   Semiconductor diode laser.

6. What happens to the order of spectrum if the distance between the particle and the screen is increased?

   The order of the spectrum decreases.

7. What will happen to the order of spectrum if the size of particle is decreased?

   The order of the spectrum decreases.

8. What is active medium?

   The medium in which population inversion is achieved is called as active medium.

9. What is population inversion?

   A state of achieving more number of atoms in the excited state than that of the ground state is called as population inversion.

10. What are the conditions required for the laser action?

    (i) Population inversion should be achieved.

    (ii) Stimulated emission should be predominant over spontaneous emission.

11. What is Pumping?

    The artificial means of achieving population inversion is called pumping.

EX.NO : 01	DETERMINATION OF WAVELENGTH OF GIVEN LASER
DATE :	

AIM :

To determine the wavelength of the given laser source using diffraction grating.

APPARATUS REQUIRED

Laser source, scale, grating, screen

FORMULA

$$(i)\ Wavelength\ of\ the\ laser\ \lambda = \frac{\sin\theta_m}{Nm}\ m$$

$$\tan\theta_m = \frac{d_m}{D} \qquad \theta_m = \tan^{-1}\left(\frac{d_m}{D}\right) degree$$

Where

N - number of lines per meter on grating (lines / m)

M - order of spectrum (No unit)

θ_m - angle of m^{th} order spectrum from $zero^{th}$ order (degrees)

m^{th} order (angle of diffraction) ($\times\ 10^{-2}$m)

d_m - distance of m^{th} order spectrum from $zero^{th}$ order.

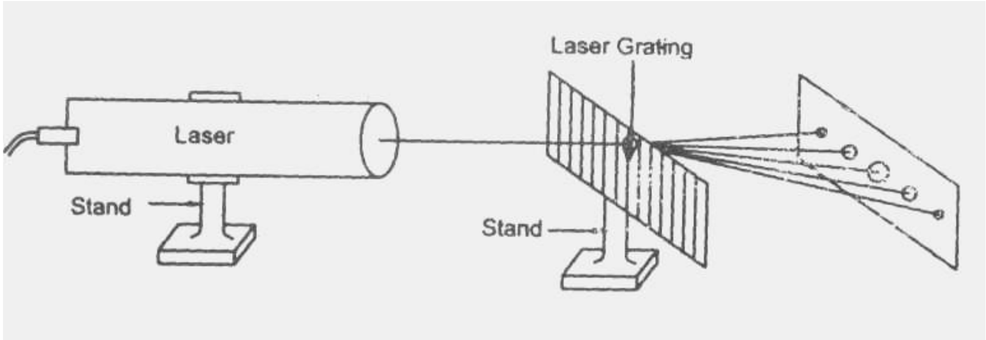
D - distance between grating and the screen. ($\times\ 10^{-2}$m)

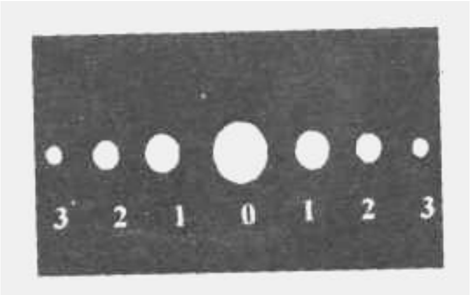
PROCEDURE

DETERMINATION OF WAVELENGTH USING GRATING

The grating is kept between laser source and the screen as shown in fig 1. Laser beam gets diffracted by the grating. Diffraction pattern of several orders is produced on the screen. The distance between the zero order and first order is noted. The experiment is repeated for different values D and d.

SCHEMATIC DIAGRAM :





OBSERVATION:

S. No.	Distance between screen and the grating (D) 10^{-2} m	Order of diffraction 'm' (No unit)	Distance between zeroth order and m^{th} order (d) $\times 10^{-2}$ m			$\theta = tan^{-1}(d/D)$ degrees	$\lambda = \frac{\sin\theta}{Nm}$ $\times 10^{-10}$ m
			Left	Right	Mean		

MEAN(λ)=

CALCULATION:

$$\text{Wavelength of the laser } \lambda = \frac{\sin\theta}{Nm} \text{ m}$$

$$\tan\theta = \frac{d}{D}$$

$$\theta = \tan^{-1}\left(\frac{d}{D}\right)$$

$$\theta = \text{.... Degree}$$

$$\lambda = \text{......} \times 10^{-10} \text{ m}$$

RESULT :

The wavelength of the laser source =-------------------------m

SAMPLE VIVA QUESTIONS

1. What is an acronym for LASER?
    a. LASER stands for Light Amplification by Stimulated Emission of Radiation.

2. What is LASER?
    a. LASER is a device which produces a highly intense monochromatic beam of light which is coherent in nature.

3. What are the properties of LASER?
4. Highly intense,
5. Coherent,
6. Directional, Monochromatic, and
7. Less divergent.

8. What is Stimulated emission?
    a. The process in which an atom an atom in the excited state returns to the ground state with an emission of emission of two photons by external stimulation.

9. What is the type of laser beam used in the experiment?
    a. Semiconductor diode laser.

10. Define Numerical Aperture
    a. It is the light gathering capacity of an optical fibre. Mathematically it is expressed as, NA = sin θ_a.

11. Define acceptance angle
    a. The maximum angle "θ_a" with which a ray of light can enter through the fibre.
12. What is the principle used in fibre optic communication system?
    a. Total internal reflection.
13. Give the condition for total internal reflection?
    (i) Light should travel from denser medium to rarer medium.
    (ii) The angle of incidence should be greater than critical angle.

12. What is Total internal reflection?

    When the angle of incidence exceeds the critical angle, the incidence ray gets reflected into the same medium. The phenomenon is called total internal reflection.

13. What happens to the order of spectrum if the distance between the particle and the screen is increased?

    The order of the spectrum decreases.

14. What will happen to the order of spectrum if the size of particle is decreased?

    The order of the spectrum decreases.

15. What is active medium?

    The medium in which population inversion is achieved is called as active medium.

16. What is population inversion?

    A state of achieving more number of atoms in the excited state than that of the ground state is called as population inversion.

17. What are the conditions required for the laser action?

    (iii) Population inversion should be achieved.

    (iv) Stimulated emission should be predominant over spontaneous emission.

18. What is Pumping?

    The artificial means of achieving population inversion is called pumping.

EX.NO : 02	DETERMINATION OF ACCEPTANCE ANGLE AND NUMERICAL APERTURE OF AN OPTICAL FIBRE
DATE :	

Aim

To determine

(i) The acceptance angle of the optical fibre.

(ii) The numerical aperture.

Apparatus Required

Diode laser, optical fibre, NA jig.

Formula

$i)\ \ Acceptance\ angle\ of\ optical\ fibre\ \theta = tan^{-1}\left(\frac{r}{d}\right)\ degree$

ii) Numerical Aperture NA =sin θ (No unit)

Where

r - radius of the circular opening in NA jig ($\times\ 10^{-2}$m)

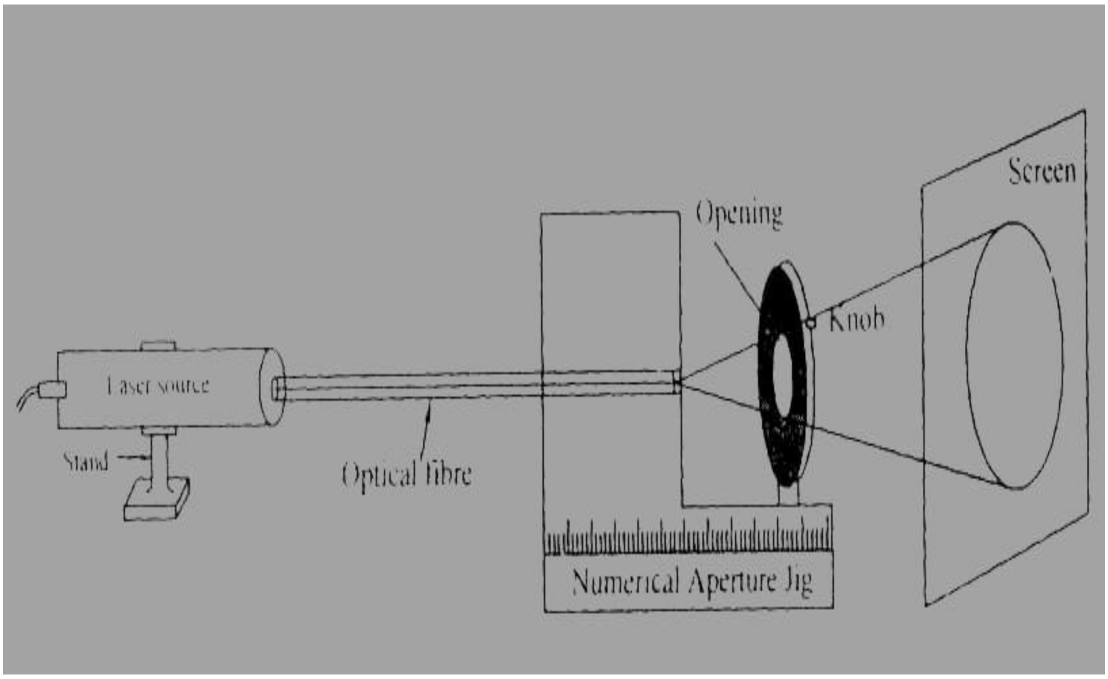
d - Distance between the tip of the optical fibre and aperture of the jig ($\times\ 10^{-2}$m)

Procedure

Determination of acceptance angle and angle of divergence

One end of optical fibre cable is connected to the laser source and the other end to the jig. The source is switched ON. The opening in the NA jig is completely opened so that a circular red patch of laser light is observed on the screen. Now the opening in the NA jig is slowly closed with the knob provided, so that a particular point the circular patch in the screen just cuts. The radius of the circular opening (r) of NA jig at which circular patch of light just cuts is measured.

SCHEMATIC DIAGRAM :



Observation

(i) Determination of acceptance angle:

S.No	Distance between Jig opening and the fibre (d) $\times 10^{-2}m$	Radius of the circular opening in Numerical aperture Jig (r) $\times 10^{-2}m$	Acceptance angle $\theta = \tan^{-1}(\frac{r}{d})$ degree
1.			
2.			
3.			
4.			
5.			
6.			
		Mean	

The distance between the NA jig opening and the fibre can be measured directly with the help of the calibration in NA jig. By substituting the values in the given formula the acceptance angle and the angle of divergence can be calculated. The same procedure can be adopted for various distances between the fibre and the opening of NA jig.

Calculation

$$(i)\ Acceptance\ angle \quad \theta = tan^{-1}(\frac{r}{d})\ degree$$

(ii) Numerical Aperture NA = $\sin\theta$

Result

(i) The acceptance angle of the optical fibre = ------------------ degree.

(ii) Numerical Aperture = --------------------- (No unit)

SAMPLE VIVA QUESTIONS

1. Ultrasonic have frequency greater than 20 KHz
2. Is Ultrasonic wave a electromagnetic wave No.
3. What type of wave is Ultrasonic wave Sound wave.
4. Give the unit of compressibility m^2N^{-1} .
5. The type of Crystal used in Ultrasonic Interferometer Piezo – electric Crystals.
6. Example for Piezo – electric Crystal Quality, tourmaline.
7. How does Velocity varies with frequency $v \times \lambda$.
8. Live the condition for maxima in acoustical grating $2d\ sin\ sin\ \theta = n\lambda$.
9. What happens to the compressibility of a liquid, when density decreases.

   Compressibility $\times$ $1/$density. ∴ when density ↓ compressibility t.
10. What is meant by acoustical Grating?

    When Ultrasonic waves are passed through the liquid due to super position of reflected and forward wave alternating stationary wave pattern is formed in a liquid. When light is passed, the liquid behaves as a diffraction grating such grating is known as Acoustical grating.
11. What is meant by compressibility?

    It is measure of the relative volume change of a fluid or solid in response to a pressure (mean stress) change.
12. What is Piezo – electric crystal?

    The crystal which exhibit Piezo – electric effect and Inverse Piezo – electric are called as Piezo – electric Crystal.

    Ex: Quartz
13. What are the various liquids that can be used in Ultrasonic interferometer?

    Water, kerosene, acetone etc.,
14. What is Piezo electric effect?

    When Piezo – electric crystal is subjected to mechanical pressure along the mechanical axis, electric charges are produced on the Electrical axis of the crystal. This phenomenon is known as Piezo – electric effect.
15. What is the relation between the wavelength of Ultrasonic wave and the number of oscillations?

$$\lambda = \frac{2d}{x}$$

EX.NO : 03	ULTRASONIC INTERFEROMETER – VELOCITY OF ULTRASONIC WAVES IN A LIQUID AND COMPRESSIBILITY OF THE LIQUID
DATE :	

Aim

(i) To determine the velocity of Ultrasonic waves in the given liquid by using ultrasonic Interferometer.
(ii) To determine the compressibility of the liquid.

Apparatus required

Ultrasonic Interferometer, Measuring cell, Frequency generator, Liquid, etc.,

Formula

(i) Velocity of Ultrasonic waves in the liquid $v = n\lambda \; m/s.$

$$Where, Wavelength\ of\ ultrasonic\ wave \quad \lambda = \frac{2d}{x}\ m$$

$$Compressibility\ of\ the\ given\ liquid\ K = \frac{1}{v^2\rho}\ m^2\ N^{-1}$$

Where, $n =$ Frequency of the generator (Hertz)
$\lambda =$ Wavelength of the ultrasonic wave (metre)
$\rho =$ Density of the liquid (kgm^{-3})
$d =$ Distance moved by the micrometer screw (metre)
$x =$ Number of oscillations (No unit)

Theory

An ultrasonic Interferometer is a device to determine the velocity of ultrasonic waves in liquid with a high accuracy. The high frequency generator generates variable frequency, which excites the quartz crystal placed at the bottom of the measuring cell (Fig.1). The excited crystal produces ultrasonic waves in the liquid. The liquid will act as an acoustical grating. Hence when ultrasonic waves passes through the rulings of grating, successive maxima and minima occurs, satisfying the condition of diffraction.

Initial adjustments:

In the high frequency generator two knobs are provided for initial adjustments. One is marked as 'Adj' the position of the needle on the ammeter is adjusted. The knob marked 'Gain' is used for the sensitivity of the instrument.

SCHEMATIC DIAGRAM:

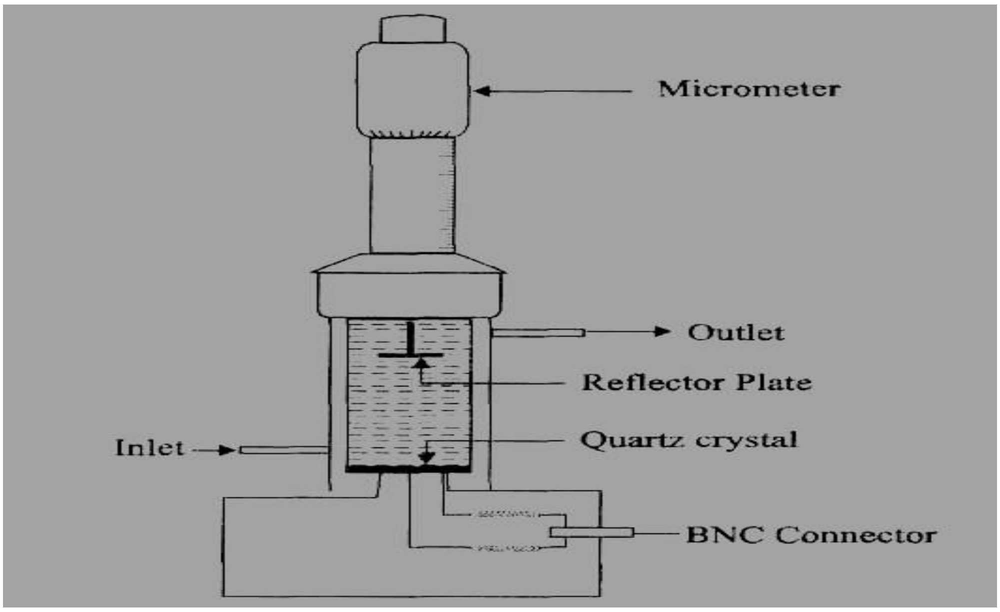


Fig 1 – Measuring Cell

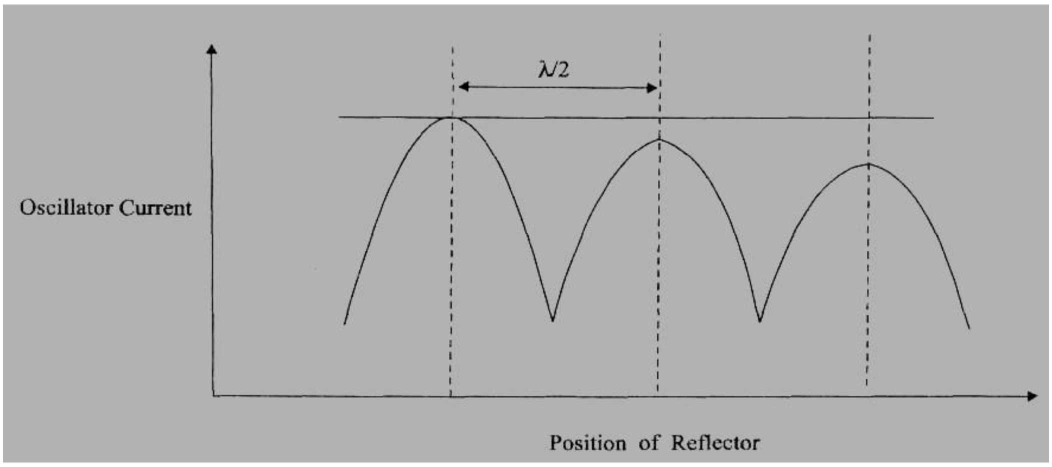


Figure 2 – Distance moved by the reflector Vs Oscillator current

Procedure

The measuring cell is connected to the output terminal of the frequency generator through a shielded cable. Liquid before switching ON the generator the cell is filled with the given. Now, when the frequency generator is switched ON the ultrasonic waves move normal from the crystal until they are reflected back by the movable reflector plane. Hence, standing waves are produced in between the reflector plate and the crystal.

The distance between the reflector plate and the crystal is changed using the micrometer screw such that the anode current increases to a maximum value and then decreases to a minimum and once becomes maximum. The distance of separation between successive maximum or minimum in the current is equal to half of the wavelength (λ) of the ultrasonic wave in the liquid medium (Fig. 2), by noting the initial and final position of the micrometer screw for one complete oscillation (maxima – minima – maxima) the distance moved (d) can be determined.

The wavelength of ultrasonic waves can be calculated using the formula $\lambda = 2d/x.$ from the value of wavelength and by noting the frequency of the generator (n), the velocity of the ultrasonic wave can be calculated using the given formula.

$$V = n\lambda$$

The compressibility of the liquid is calculated using the given formula.

$$K = \frac{1}{v^2 \rho}\ m^2\ N^{-1}$$

OBSERVATION:

S.No.	No. of Oscillations (x) (No Unit)	Readings for "x" Oscillations — Initial Reading: PSR $\times 10^{-3}$m	HSC Div	TR $\times 10^{-3}$m	Distance $\times 10^{-3}$ m	Wavelength $\lambda = \frac{2d}{x}$ $\times 10^{-3}$ m	Velocity $v = n\lambda$ $\times 10^{3}$ m/s

Calculation:

To find the Compressibility of liquid:

Velocity of Ultrasonic wave in the given liquid $v = n\lambda\ ms^{-1}$

$$Wavelength\ \lambda = \frac{2d}{x} m$$

$$Compressibility\ of\ the\ given\ liquid\ K = \frac{1}{v^2\rho}\ m^2\ N^{-1}$$

$\rho =$_____________kgm^3

$x =$ _____________

v =______________ms^{-1}

Result

(i) The velocity of Ultrasonic waves in the liquid =............................ $\times 10^3\ ms^{-1}$

ii) Compressibility of the liquid = ... $\times 10^{-10}\ m^2N^{-1}$

SAMPLE VIVA QUESTIONS

1. Name any four bad conductors.
   Glass, paper, rubber, cardboard, ebonite.

2. What will happen to the thermal conductivity, if the thickness of the bad conductor is increased?
   It increases.

3. Will there be any change in conductivity if the area of the bad conductor is decreased?
   Yes the diameter of the bad conductor should match the diameter of the disc otherwise the steady state cannot be reached.

4. Should the diameter of the bad conductor match the diameter of the disc? Why?
   Yes, the heat transfer will be uniform only when the diameter matches.

5. Is there any reason to take the specimen in the form of a disc?
   A thin disc is taken because its area of cross section is large while thickness is small, it increases the quantity of heat conducted across its faces.

6. Give the unit of thermal conductivity.
   W/mk

7. Give the unit of specific heat capacity.
   J/Kg K

8. What is meant by rate of cooling?
   The variation or decrease in temperature with respect to time is called rate of cooling. i.e. $d\theta/dt$

9. Define thermal conductivity? Give its unit.
   It is defined as the quantity of heat conducted per second normally across unit area of cross – section of the material per unit temperature difference. It denotes the heat conducting power. Its unit is Watt $metre^{-1}\ Kelvin^{-1}$

10. Can this method is used for good conductors?
    No, in that case, due to large conduction of heat, the temperature recorded by T_1 and T_2 will be very nearly the same.

EX.NO : 04	THERMAL CONDUCTIVITY OF THE BAD CONDUCTOR USING LEE'S DISC METHOD
DATE :	

Aim

To determine the thermal conductivity of a bad conductor using Lee's disc, method

Apparatus Required

Lee's disc apparatus, steam boiler, thermometers, bad conductor, screw gauge, stop clock etc.,

Formula

Thermal conductivity of the bad conductor

$$K = \frac{MSRd(r + 2h)}{\pi r^2(\theta_1 - \theta_2)(2r + 2h)} \frac{W}{m} K$$

Where

M – Mass of metallic (brass) disc $(\times 10^{-3} kg)$.

S – Specific heat capacity of the metallic disc $(\frac{J}{kg} K)$

$$R = \frac{d\theta}{dt} - Rate\ of\ cooling\ at\ stady\ state\ temperature\ \theta_2\ (\frac{℃}{second})$$

$calclated\ (from\ graph)$.

$d\theta$ – Change in temperature $(\,^oC\,)$

dt – Change in time $(second)$

d – Thickness of the bad conductor disc $(\times 10^{-3} m)$

r – Radius of the metallic disc $(\times 10^{-2} m)$

h – Thickness of the metallic disc $(\times 10^{-3} m)$

θ_1 – Steady state temperature of the steam chamber $(\,^oC\,)$

θ_2 – Steady state temperature of the metallic disc $(\,^oC\,)$

Observation

(i) To find the thickness of the brass disc (h) using screw gauge

ZE = division

LC = 0.01 mm ZC = mm

S.No	PSR mm	HSC division	OR = PSR + (HSC $\times$ LC) mm	CR = OR $\pm$ ZC mm
1.				
2.				
3.				
4.				
5.				
			Mean	

(ii) To find the thickness of the cardboard (h) using screw gauge

ZE = division

LC = 0.01 mm ZC = mm

S.No	PSR mm	HSC division	OR = PSR + (HSC $\times$ LC) mm	CR = OR $\pm$ ZC mm
1.				
2.				
3.				
4.				
5.				
			Mean	

SCHEMATIC DIAGRAM:

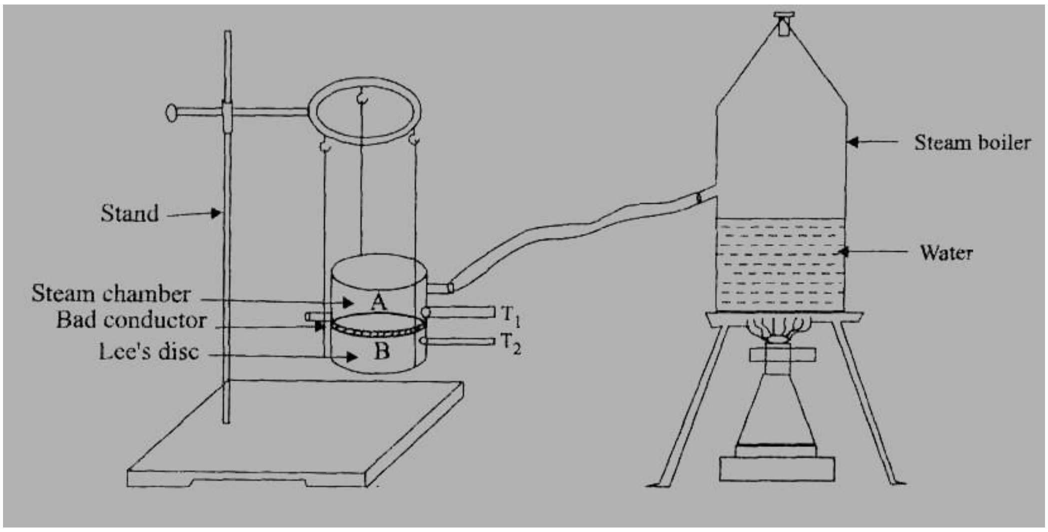


Figure 1 Lee's Disc arrangement

Model Graph

R – By Graphical Method

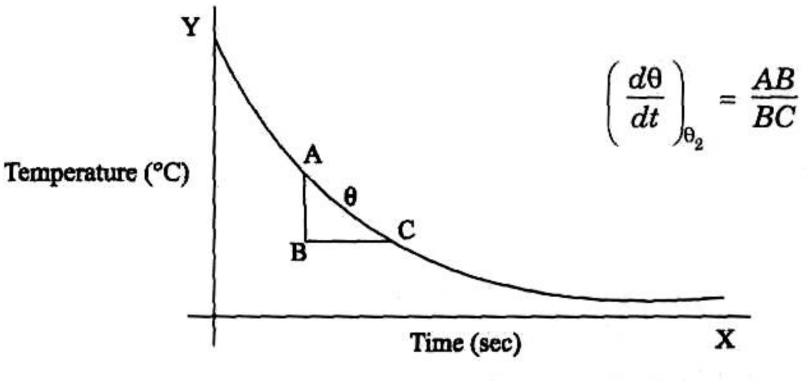


Figure 2

Procedure

Initially, the thickness d of the cardboard and thickness of brass disc (h) is measured using screw gauge. The mass M of the brass disc is noted using a biscuit balance. The circumference of the disc is found using thread and the radius is calculated. The given bad conductor is placed in between the metallic disc and steam chamber. Two thermometers T_1 *and* T_2 are inserted in the respective holes.

(a) Determination of steady state temperature θ_2

Steam is passed through the steam chamber for a sufficiently long time until thermometers T_1 *and* T_2 indicates steady temperatures θ_1 *and* θ_2 respectively.

(b) Determination of rate of cooling (R) of the brass disc

$$R = \frac{d\theta}{dt} = \frac{Change\ in\ temperature}{Change\ in\ time}\ °C/Second$$

After noting the steady temperature θ_2°C, the bad conductor disc is removed. The brass disc is heated in direct contact with the steam chamber until the temperature rises by about 10°C above the steady temperature θ_2. The steam chamber is carefully removed after cutting the steam flow. When the temperature of the disc reaches 5°C above the steady state temperature $\theta_2 + 5$°C of the disc, a stop clock is ON. Time for every 1°C fall temperature is noted until the metallic disc attains the temperature($\theta_2 -$ 5°C). A graph is drawn taking time t along X – axis and temperature T along Y – axis. The cooling curve is obtained the slope AB / BC gives the rate of cooling $R = \frac{d\theta}{dt}$ as shown in figure (2).

$$(iii)\ To\ find\ rate\ of\ cooling\ \frac{d\theta}{dt} = R$$

$\theta_1 =$ ………….. $\theta_2 =$ …………..

Temperature (°C)	Time (Second)

Calculations

Mass of the brass disc M = ---------------------$\times 10^{-3} Kg$

Thickness of the brass disc h = --------------------- $\times 10^{-3} m$

Thickness of the bad conductor disc d = --------------------- $\times 10^{-3} m$

Radius of the bad conductor disc r =--------------------- $\times 10^{-2} m$

Specific heat capacity of the material of the brass disc

S = 372 J/Kg/K

Steady state temperature of the steam chamber $\theta_1 =$ ---------------------℃

Steady state temperature of the brass disc $\theta_2 =$ --------------------- ℃

Rate of cooling at the steady state temperature R =--------------------- $℃/Sec$

(From graph)

$$K = \frac{MSRd(r + 2h)}{\pi r^2(\theta_1 - \theta_2)(2r + 2h)} \frac{W}{m} K$$

Result

Thermal conductivity of bad conductor using Lees's disc method K= ------------------ W / m K

SAMPLE VIVA QUESTIONS

1. What is meant by viscosity? Give its unit.
   Viscosity is defined as the property of a liquid in which the tangential force the drags the liquid when it flows on a solid surface.

2. Define Co–efficient of viscosity.
   The tangential force acting per unit area over two adjacent layers of the liquid for a unit velocity gradient is known as coefficient of viscosity.

3. State the relation between density and viscosity.
   Viscosity is directly proportional to the density of the liquid.

4. How do you determine the radius of the capillary tube?
   The horizontal cross wire of the microscope is focused to the circumference of the bore of the bore of the capillary tube at the top and bottom position. The difference between the two readings gives the diameter from which the radius can be calculated.

5. Give the difference between stream line flow and turbulent flow.
   In streamline flow the liquid flow with constant velocity at all points.
   In turbulent flow, the liquid moves with different velocities in different velocities in different direction.

6. Give the value of viscosity of water and kerosene at room temperature.
   Water - $8 \times 10^{-4}\ Nsm^{-1}$
   Kerosene - $2 \times 10^{-3}\ Nsm^{-1}$

7. How height “h” is determined?
   The value of height “h” is determined by finding the average of the height for the initial and final position in the burette from the surface of the table.

8. Give the relation between pressure and viscosity.
   Viscosity is directly proportional to the pressure of the liquid.

9. Is the flow through capillary tube stream lined or turbulent motion.
   The flow is streamlined.

10. Can you use this method for all types of liquids?
    No. This method can be used only for low viscous liquids.

EX.NO : 05	COEFFICIENT OF VISCOSITY – POISEUILLE'S FLOW METHOD
DATE :	

Aim

To determine the coefficient of viscosity of the given liquid by Poiseuille's flow method.

Apparatus Required

Burette, a stand, capillary tube, beaker, stop clock, travelling microscope, liquid, rubber tube.

Formula

$$Coefficient\ of\ viscosity\ of\ the\ given\ liquid,$$

$$\eta = \frac{\pi\rho\ g a^4\ ht}{8\ l V}\ Nsm^{-2}$$

Where

ρ = Density of the given liquid ($kg\ m^{-3}$)

g = Acceleration due to gravity (ms^{-2})

a = Radius of the capillary tube ($\times 10^{-2}\ m$)

h = Mean pressure head ($\times 10^{-2}\ m$)

t = Time of flow (second)

l = Length of capillary tube ($\times 10^{-2}\ m$)

V = Volume of the liquid ($\times 10^{-6} m^3$)

Principle

The liquid is allowed to flow through a uniform capillary tube which is held horizontally under constant pressure difference between the two ends of the capillary. The flow of liquid through the tube is streamline and the layers which are in contact of the walls of the tube are at rest. The top most layer have the maximum velocity. The intermediate layers have intermediate velocity. To maintain this relative motion of the layers, an external force must be acting on the liquid. Otherwise the liquid will come to rest due to internal frictional forces acting between the layers of the liquid. These internal frictional forces that bring the liquid to rest are known as viscous force and this property is known as viscosity.

The property by virtue of which the relative motion between the layers of a liquid is maintained is called viscosity. We can also say viscosity is the resistance to flow.

SCHEMATIC DIAGRAM:

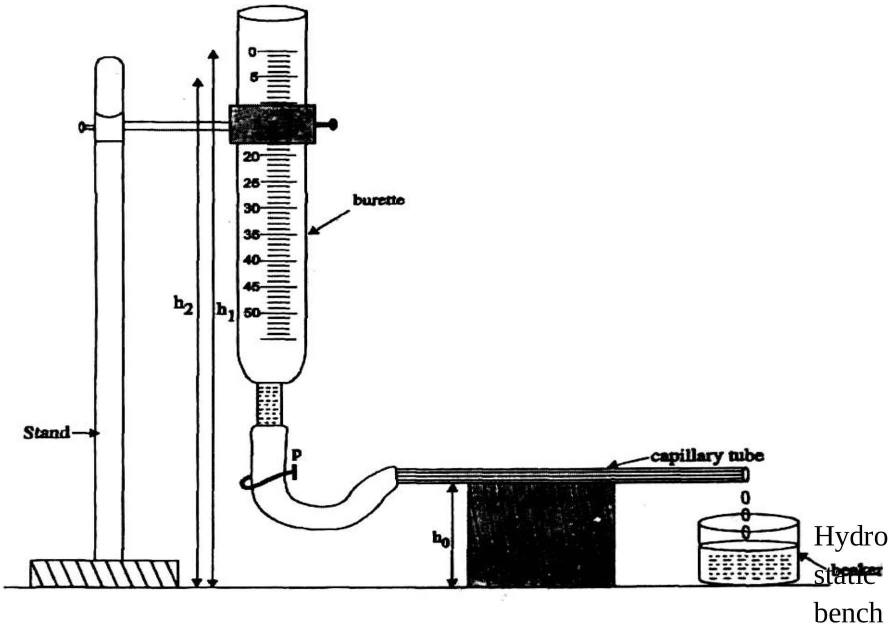


Figure 1 – Poiseuille's flow set - up

OBSERVATION: Height of Capillary tube from Table h_0

S.No	Water level in the Burette (cc)	Time (t) (second)	Range (cc)	Time for flow of 10 cc liquid (second)	Height of initial water level from table (h_1) x 10^{-2} m	Height of initial water level from table (h_2) x 10^{-2} m	Pressure head $h = \frac{h_1 + h_2}{2} - h_3$	ht x 10^{-2} m

Procedure

A graduated burette is mounted vertically in the stand. A rubber tube is connected to the bottom of the burette. To the other end of the tube, a capillary tube is inserted and placed in a perfectly horizontal position.

The pure water is poured in to the burette using a funnel. The height of the liquid level in the burette is adjusted such that there is a streamline flow of water through the capillary tube i.e)5 to 8 drops per minute. This can be achieved by raising or lowering the capillary tube. When the water level in the burette comes to zero mark a stopwatch is started and the time for the water level to reach 5cc, 10cc, 15cc, 20cc and 25cc mark is noted and entered in the tabular column. From the above readings, the time of flow of water (t) 5cc, at various ranges like 0cc to 5cc, 5cc to 10cc, 10cc to 15cc, 15cc to 20cc, etc., are tabulated. In each case the heights of the respective initial levels and final levels from the experimental table are noted. The average pressure head height is calculated and entered in the tabular column. Substituting the average value of ht in the given formula, the coefficient of viscosity of water is calculated.

To find the Radius of the capillary tube:

The travelling microscope is used to measure the radius of the capillary tube (a) .The capillary tube is fixed on the stand and the travelling microscope is adjusted to view the inner circle of the capillary tube. The vertical cross wire is adjusted to coincide with the left side (A_1) of the inner circle. The corresponding MSR and VSC are noted. Similarly the cross wire is adjusted with the right side (A_2) of the inner circle and the readings are noted.The same procedure is repeated using the horizontal cross wire of the telescope and the corresponding readings B_1 and B_2 are tabulated. The inner diameter of the capillary tube is calculated by finding the difference between A_1 and A_2 , B_1 and B_2.The average value of diameter is used for the calculation.

Calculations

Density of the given liquid ρ = $1000 kg\ m^{-3}$

Acceleration due to gravity g = $9.8 ms^{-2}$

Radius of the capillary tube a = ------------ $\times 10^{-2}\ m$

Length of capillary tube l = ------------ $\times 10^{-2}\ m$

Volume of the liquid V = ------------ $\times 10^{-6} m^3$

ht = ------------ $\times 10^{-2}\ m\ s$

The viscosity of the given liquid is,

$$\eta = \frac{\pi \rho g a^4}{8\,l} \left(\frac{ht}{V}\right) Nsm^{-2}$$

Result

The coefficient of viscosity of the liquid, η = -------------------- Nsm^{-2}